LAB WORKSHEET

Database Design Lab

Normalise a paper form to third normal form, write a data dictionary, generate SQL and draw an ERD with the right cardinality.

Key words

- **Primary key**: uniquely identifies a row. **Composite key**: two or more columns together. **Foreign key**: holds another table's primary key.
- **Functional dependency** A → B: know A and there is only one B. **Partial**: depends on part of a composite key. **Transitive**: depends on another non-key column.
- **Anomalies**: update (same fact stored many times), insert (can't add a fact without another), delete (removing a row loses another fact).

The steps

1. **1NF**: remove the repeating group (one item per row) and choose the key — usually a composite key.
2. **2NF**: "the whole key" — move columns that depend on part of the key into their own table.
3. **3NF**: "nothing but the key" — move columns that depend on another non-key column into a table keyed by that column; leave it behind as a foreign key.

Data dictionary

For every field: data type (Integer, Real, Currency, Text, Date, Time, Boolean), size, key (PK, FK, PK + FK) and validation (presence, length, range, format, type, list, lookup). Phone numbers and codes are Text. Foreign keys get a lookup check (referential integrity).

ERDs and cardinality

- Every foreign key is a relationship. The table holding the foreign key is on the **many** side.
- UML multiplicities: 1, 0..1, 1..*, 0..* — the value next to an entity says how many of that entity link to one of the other.
- Say it both ways: "Each Customer places zero or more Orders. Each Order is placed by exactly one Customer."
- Many-to-many becomes a link table with two one-to-many relationships.

Independent task

Design a database for a fictional organisation from its paper form with no step-by-step help: a correct 3NF design on the workbench, a correct data dictionary, then a correct ERD.

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